Lab 7 - Install Cura and Slice Your First Model

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 02: Process Related Issues for AM TSC objective: A2 - set up the slicing toolchain used to analyse and fix AM process issues

Goal

Install Ultimaker Cura, add your printer profile, and slice a calibration cube to understand how the slicer turns an STL into G-code printer instructions.

What you'll build

A working Cura installation with your printer profile, and your first sliced G-code file.

Tools: Ultimaker Cura, cube10.stl, cube20.stl

Steps

1. Download Ultimaker Cura for your operating system and run the installer.

```
https://ultimaker.com/software/ultimaker-cura
```

2. Launch Cura and step through the first-run wizard; skip account sign-in if offered.

```
Cura > Get Started
```

3. Add your printer - pick the exact model, or 'Custom FFF printer' with your bed size and nozzle diameter.

```
Settings > Printer > Add Printer
```

4. Open the sample cube from the lab folder.

```
File > Open File(s) > cube10.stl
```

5. Select the Standard Quality 0.2 mm profile and note the default wall, infill, speed and temperature values.

```
Print Settings > Profile - Standard Quality 0.2 mm
```

6. Slice the model and read the estimated print time and material use.

```
Slice > note time + grams
```

7. Open the Preview tab and drag the layer slider to watch the toolpath layer by layer - walls, infill, travels.

```
Preview > layer slider
```

8. Save the G-code to disk (or SD card) - this file is the printer's complete instruction set.

```
Save to Disk > cube10.gcode
```

9. Repeat the slice with cube20.stl, confirm Cura reads it as 20 × 20 × 20 mm, and compare the time/material estimate with cube10.stl.

```
File > Open File(s) > cube20.stl > Scale > verify 20 mm on X/Y/Z
```

Evidence to capture

- Screenshot of cube20.stl in Cura Preview with walls, infill and travel moves visible.
- Recorded print-time and material estimates for both the 10 mm and 20 mm cubes.

Test it

Cura shows per-layer toolpaths for both cubes, cube20.stl measures 20 × 20 × 20 mm, and your comparison explains why doubling linear size increases print time and material by much more than two times.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.
- cube20.stl - 20 mm calibration cube for scale, dimensional and time/material comparisons.